Looping Statement - II

Session Objectives

- ✅ Understand what looping statements are
- ✅ Understand what a for loop is
- ✅ Shorthand for loop (List Comprehension)
- ✅ Understand is nested for/while loop
- ✅ Understand what a while loop is

```
# max_element
num_list = [55,77,91,99,-11,False,True,121,71,82,33,41,90]
max_val = float('-inf')
for val in num_list:
    if val > max_val:
        max_val = val

max_val
```

```
121
```

```
# max_element
num_list = [55,77,91,99,-11,False,True,121,71,82,33,41,90]
max_val = num_list[0] # 55
for val in num_list:
    if val > max_val:
        max_val = val

max_val
```

```
121
```

```
# min_element
num_list = [55,77,91,99,-11,False,True,121,71,82,33,41,90]
min_val = float('inf')
for val in num_list:
    if val < min_val:
        min_val = val

min_val # -11
```

```
-11
```

```
# min_element
num_list = [55,77,91,99,-11,False,True,121,71,82,33,41,90]
min_val = num_list[0]
for val in num_list:
```

```python
min_val = num_list[0]
for val in num_list:
    if val < min_val:
        min_val = val

min_val # -11
```

```
-11
```

```python
# Looping Statement
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor']
print(car_list[0])
print(car_list[1])
print(car_list[2])
print(car_list[3])
print(car_list[4])
print(car_list[5])
print(car_list[6])
print(car_list[7])
print(car_list[8])
```

```
Taigun
Creta
Thar
Slavia
Scorpio
Hummer
Verna
Defender
Kylaq
```

```python
for car in car_list:
    print(car)
```

```
Taigun
Creta
Thar
Slavia
Scorpio
Hummer
Verna
Defender
Kylaq
Astor
```

```python
# .append() -> Adding an element to the end of a list
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor']
fav_car_list = []
for car in car_list:
    if (car == 'Slavia') or (car == 'Hummer') or (car == 'Defender'):
        fav_car_list.append(car)

fav_car_list
```

```
['Slavia', 'Hummer', 'Defender']
```

```python
# .append() -> Adding an element to the end of a list ['Dynamic']
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor']
fav_car_list = []
```

```python
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor']
fav_car_list = []

fav_car1 = input("Enter your 1st Fav Car: ")
fav_car2 = input("Enter your 2nd Fav Car: ")
fav_car3 = input("Enter your 3rd Fav Car: ")

for car in car_list:
    if (car == fav_car1) or (car == fav_car2) or (car == fav_car3):
        fav_car_list.append(car)

fav_car_list
```

```
Enter your 1st Fav Car:  Fortuner
Enter your 2nd Fav Car:  Thar
Enter your 3rd Fav Car:  Virtus
['Thar']
```

```python
# .append() -> Adding an element to the end of a list ['Dynamic']
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

car_set = set()
for car in car_list:
    car_set.add(car) # [Unordered, Unique]

print(car_set)
```

```
{'Thar', 'Kushaq', 'Kylaq', 'Alto', 'Defender', 'Taigun', 'Creta', 'Slavia', 'Carens', 'Fortuner', 'Hummer',
'Virtus', 'Nano', 'Scorpio', 'Verna', 'Astor', 'Brezza'}
```

```python
# String Iteration -> A sequence of characters ~ String
_str = "Python Programming"
for char in _str:
    print(char , end = " ")
```

```
P y t h o n   P r o g r a m m i n g
```

```python
# String Iteration -> A sequence of characters ~ String
_str = "Python Programming"
char_list = []
for char in _str:
    char_list.append(char)

print(char_list)
```

```
['P', 'y', 't', 'h', 'o', 'n', ' ', 'P', 'r', 'o', 'g', 'r', 'a', 'm', 'm', 'i', 'n', 'g']
```

```python
# slicing[Start = 0, Stop : [Length] (Non-Inclusive) , Step = +1]
# range(start,stop,step)
for i in range(10): # [0,1,2...9]
    print(i , end = " ")
```

```
0 1 2 3 4 5 6 7 8 9
```

```python
for i in range(1,11): # [1,2...9,10]
    print(i , end = " ")
```

```
1 2 3 4 5 6 7 8 9 10
```

```python
# Odd Elements
for i in range(1,11,2): # [1,3,5,7,9]
    print(i , end = " ")
```

```
1 3 5 7 9
```

```
    print(i , end = " ")
```

```
1 3 5 7 9
```

```
# Even Elements
for i in range(0,11,2): # [0,2,4,6,8,10]
    print(i , end = " ")
```

```
0 2 4 6 8 10
```

```
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

for car in car_list:
    print(car , end = " ")
```

```
Taigun Creta Thar Slavia Scorpio Hummer Verna Defender Kylaq Astor Fortuner Carens Virtus Verna Scorpio Alto T
har Creta Nano Brezza Kushaq
```

```
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

stop = len(car_list)
print(stop) # 21
for car in range(stop): # [0,1,2....20]
    print(car , end = ": ")
    print(car_list[car])
```

```
21
0: Taigun
1: Creta
2: Thar
3: Slavia
4: Scorpio
5: Hummer
6: Verna
7: Defender
8: Kylaq
9: Astor
10: Fortuner
11: Carens
12: Virtus
13: Verna
14: Scorpio
15: Alto
16: Thar
17: Creta
18: Nano
19: Brezza
20: Kushaq
```

```
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

stop = len(car_list)
print(stop) # 21
new_car_list = []
for car in range(stop): # [0,1,2....20]
    print(car , end = " ")
    new_car_list.append(car_list[car])

print()
```

```python
    new_car_list.append(car_list[car])

print()
print(new_car_list)
```

```
21
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
['Taigun', 'Creta', 'Thar', 'Slavia', 'Scorpio', 'Hummer', 'Verna', 'Defender', 'Kylaq', 'Astor', 'Fortuner',
'Carens', 'Virtus', 'Verna', 'Scorpio', 'Alto', 'Thar', 'Creta', 'Nano', 'Brezza', 'Kushaq']
```

```python
# Filling an alternative elements from a list
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

stop = len(car_list)
print(stop) # 21
alt_car_list = []
for car in range(0,stop,2): # [0,2,4,6....20]
    print(car , end = " ")
    alt_car_list.append(car_list[car])

print()
print(alt_car_list)
```

```
21
0 2 4 6 8 10 12 14 16 18 20
['Taigun', 'Thar', 'Scorpio', 'Verna', 'Kylaq', 'Fortuner', 'Virtus', 'Scorpio', 'Thar', 'Nano', 'Kushaq']
```

```python
# Filling up the reverse_car_list
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

stop = len(car_list)
print(stop) # 21
reverse_car_list = []
for car in range(stop-1,-1,-1): # [20,19,18.....0]
    print(car , end = " ")
    reverse_car_list.append(car_list[car])

print()
print(reverse_car_list)
```

```
21
20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
['Kushaq', 'Brezza', 'Nano', 'Creta', 'Thar', 'Alto', 'Scorpio', 'Verna', 'Virtus', 'Carens', 'Fortuner', 'Ast
or', 'Kylaq', 'Defender', 'Verna', 'Hummer', 'Scorpio', 'Slavia', 'Thar', 'Creta', 'Taigun']
```

```python
# Loop with Enumerate() -> (index , value)
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

for index , car_name in enumerate(car_list):
    print(f"{index}:  {car_name}")
```

```
0:  Taigun
1:  Creta
2:  Thar
3:  Slavia
4:  Scorpio
5:  Hummer
6:  Verna
7:  Defender
8:  Kylaq
9:  Astor
```

```
7:  Defender
8:  Kylaq
9:  Astor
10:  Fortuner
11:  Carens
12:  Virtus
13:  Verna
14:  Scorpio
15:  Alto
16:  Thar
17:  Creta
18:  Nano
19:  Brezza
20:  Kushaq
```

```python
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

for index , car_name in enumerate(car_list): # (idx,val)
    print(car_list[index] , end = " ")
```

```
Taigun Creta Thar Slavia Scorpio Hummer Verna Defender Kylaq Astor Fortuner Carens Virtus Verna Scorpio Alto T
har Creta Nano Brezza Kushaq
```

```python
# Enumerate returns the index and the values/elements of an iterables.
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
            'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']

for index , car_name in enumerate(car_list): # (idx,val)
    print(car_name, end = " ")
```

```
Taigun Creta Thar Slavia Scorpio Hummer Verna Defender Kylaq Astor Fortuner Carens Virtus Verna Scorpio Alto T
har Creta Nano Brezza Kushaq
```

```python
_dict = {
    'name' : 'Paras',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Gurugram',
    'state' : 'Haryana',
    'country' : 'India'
}
_dict.keys()
```

```
dict_keys(['name', 'age', 'gender', 'city', 'state', 'country'])
```

```python
_dict.values()
```

```
dict_values(['Paras', 29, 'Male', 'Gurugram', 'Haryana', 'India'])
```

```python
_dict.items()
```

```
dict_items([('name', 'Paras'), ('age', 29), ('gender', 'Male'), ('city', 'Gurugram'), ('state', 'Haryana'),
('country', 'India')])
```

```python
for key,value in _dict.items():
    print(f"{key} : {value}")
```

```
name : Paras
age : 29
gender : Male
city : Gurugram
state : Haryana
country : India
```

```python
_dict
```

```
state : Haryana
country : India
```

```
_dict
```

```
{'name': 'Paras',
 'age': 29,
 'gender': 'Male',
 'city': 'Gurugram',
 'state': 'Haryana',
 'country': 'India'}
```

```python
# Else with For Loops: [Synchronized Code]
for x in range(1,11): #[1,2,3,..10]
    print(x , end = " ")
else:
    print()
    print("Loop Run Successfully ✅")
```

```
1 2 3 4 5 6 7 8 9 10
Loop Run Successfully ✅
```

```python
# Break Statement -> Tries to break the Loop Abruptly
for x in range(1,11): #[1,2,3,..10]
    print(x , end = " ")
    if x == 7:
        break
else:
    print()
    print("Loop Run Successfully ✅")
```

```
1 2 3 4 5 6 7
```

```python
# Print all the factors of 'n'
n = int(input("Enter the Value of 'n': "))
for i in range(2,n): # [2,3,4....n-1]
    if n % i == 0:
        print(i , end = " ")
```

```
Enter the Value of 'n':  15
3 5
```

Memory

```
n = 15
i = [2,3,4,5...14]
15 % 2 == 0 : False
15 % 3 == 0 : True
15 % 4 == 0 : False
15 % 5 == 0 : True
```

```python
# Print all the factors of 'n'
n = int(input("Enter the Value of 'n': "))
for i in range(2,n): # [2,3,4....n-1]
    if n % i == 0:
        print(i , end = " ")
```

```
Enter the Value of 'n':  64
2 4 8 16 32
```

```python
# Print all the factors of 'n'
n = int(input("Enter the Value of 'n': "))
for i in range(2,n//2+1):
```

```
# Print all the factors of 'n'
n = int(input("Enter the Value of 'n': "))
for i in range(2,n//2+1):
    print(i , end = " ")
```

```
Enter the Value of 'n':  64
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
```

```
# Print all the factors of 'n'
n = int(input("Enter the Value of 'n': "))
for i in range(2,n//2+1):
    if n % i == 0:
        print(i , end = " ")
```

```
Enter the Value of 'n':  64
2 4 8 16 32
```

```
new_list = [expression for item in iterable (optional condition)]
or
new_list = [expression1 if condition else expression2 for item in iterable]
```

Syntax : Shorthand for loop (List Comprehension)

```
new_list = [expression for item in iterable (optional condition)]
or
new_list = [expression1 if condition else expression2 for item in iterable]
```

```
# new_list = [expression for item in iterable (optional condition)]
num_list = [1,2,3,4,5,6,7,8,9,10,11]
squared = [x**2 for x in num_list]
squared
```

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121]
```

```
num_list = [1,2,3,4,5,6,7,8,9,10,11]
add_10 = [x+10 for x in num_list]
add_10
```

```
[11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21]
```

```
# Generate the even_list from an existing num_list
num_list = [1,2,3,4,5,6,7,8,9,10,11]
even_list = [x for x in num_list if (x % 2 == 0)]
even_list
```

```
[2, 4, 6, 8, 10]
```

```
# Long Hand Approach
num_list = [1,2,3,4,5,6,7,8,9,10,11]
even_list = []
for val in num_list:
    if val % 2 == 0:
        even_list.append(val)
even_list
```

```
[2, 4, 6, 8, 10]
```

```
# Generate the odd_list from an existing num_list
num_list = [1,2,3,4,5,6,7,8,9,10,11]
odd_list = [x for x in num_list if (x % 2 != 0)]
```

```python
# Generate the odd_list from an existing num_list
num_list = [1,2,3,4,5,6,7,8,9,10,11]
odd_list = [x for x in num_list if (x % 2 != 0)]
odd_list
```

```
[1, 3, 5, 7, 9, 11]
```

```python
# Long Hand Approach
num_list = [1,2,3,4,5,6,7,8,9,10,11]
odd_list = []
for val in num_list:
    if val % 2 == 1:
        odd_list.append(val)
odd_list
```

```
[1, 3, 5, 7, 9, 11]
```

```python
# Replacing the elements conditionally
# new_list = [expression1 if condition else expression2 for item in iterable]
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
           'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']
new_car_list = [car if car != 'Thar' else "Curvv"  for car in car_list]
print(new_car_list , end = " ")
```

```
['Taigun', 'Creta', 'Curvv', 'Slavia', 'Scorpio', 'Hummer', 'Verna', 'Defender', 'Kylaq', 'Astor', 'Fortuner',
'Carens', 'Virtus', 'Verna', 'Scorpio', 'Alto', 'Curvv', 'Creta', 'Nano', 'Brezza', 'Kushaq']
```

```python
# Replacing the elements conditionally
# Long Hand Approach
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor',
           'Fortuner','Carens','Virtus','Verna','Scorpio','Alto','Thar','Creta','Nano','Brezza','Kushaq']
new_car_list = []
for car in car_list:
    if car != 'Thar':
        new_car_list.append(car)
    else:
        new_car_list.append("Curvv")
print(new_car_list , end = " ")
```

```
['Taigun', 'Creta', 'Curvv', 'Slavia', 'Scorpio', 'Hummer', 'Verna', 'Defender', 'Kylaq', 'Astor', 'Fortuner',
'Carens', 'Virtus', 'Verna', 'Scorpio', 'Alto', 'Curvv', 'Creta', 'Nano', 'Brezza', 'Kushaq']
```

```python
# 2D Pattern in List Comprehension
[(i,j) for i in range(4) for j in range(4)]
```

```
[(0, 0),
 (0, 1),
 (0, 2),
 (0, 3),
 (1, 0),
 (1, 1),
 (1, 2),
 (1, 3),
 (2, 0),
 (2, 1),
 (2, 2),
 (2, 3),
 (3, 0),
 (3, 1),
 (3, 2),
 (3, 3)]
```

```python
# Diagonal Matrix (i == j)
```

```
(3, 3)]
```

```python
# Diagonal Matrix (i == j)
[(i,j) for i in range(4) for j in range(4) if (i==j)]
```

```
[(0, 0), (1, 1), (2, 2), (3, 3)]
```

```python
# Nested For Loop:
for i in range(4):
    for j in range(4):
        print((i,j) , end = " ")
    print()
```

```
(0, 0) (0, 1) (0, 2) (0, 3)
(1, 0) (1, 1) (1, 2) (1, 3)
(2, 0) (2, 1) (2, 2) (2, 3)
(3, 0) (3, 1) (3, 2) (3, 3)
```

```python
# Nested For Loop:
for i in range(4):
    for j in range(4):
        if i == j: # Diagonal Elements
            print((i,j) , end = " ")
    print()
```

```
(0, 0)
(1, 1)
(2, 2)
(3, 3)
```

```python
# Nested For Loop:
for i in range(4):
    for j in range(4):
        print((i,j) , end = " ")
    print()
```

```
(0, 0) (0, 1) (0, 2) (0, 3)
(1, 0) (1, 1) (1, 2) (1, 3)
(2, 0) (2, 1) (2, 2) (2, 3)
(3, 0) (3, 1) (3, 2) (3, 3)
```

Memory

```
i = [0,1,2,3]
j = [0,1,2,3]
```

i = [0,1,2,3]

j = [0,1,2,3]

i == j

(0,0)	(0,1)	(0,2)	(0,3)
(1,0)	(1,1)	(1,2)	(1,3)
(2,0)	(2,1)	(2,2)	(2,3)
(3,0)	(3,1)	(3,2)	(3,3)